NLP Lab 7: Part-of-Speech Tagging for Hindi

Naive Bayes and Hidden Markov Model taggers with forward, backward, and forward-backward Viterbi decoding

1. Data

The corpus is the Hindi POS-tagged data supplied with the lab (word/tag pairs, one per line, blank lines between sentences). The train split has 15,087 sentences and 325,710 tokens over 24 tags. Dev has 1,866 sentences and test has 1,910.

2. Methods

Naive Bayes. Each token is classified independently from five categorical features: a binary length feature (MORE if the word has more than 4 characters, else LESS), the word, the previous word, the next word, and the previous tag. Features use Laplace smoothing (alpha = 1). Training uses gold previous tags; prediction is greedy left-to-right, using the predicted previous tag.

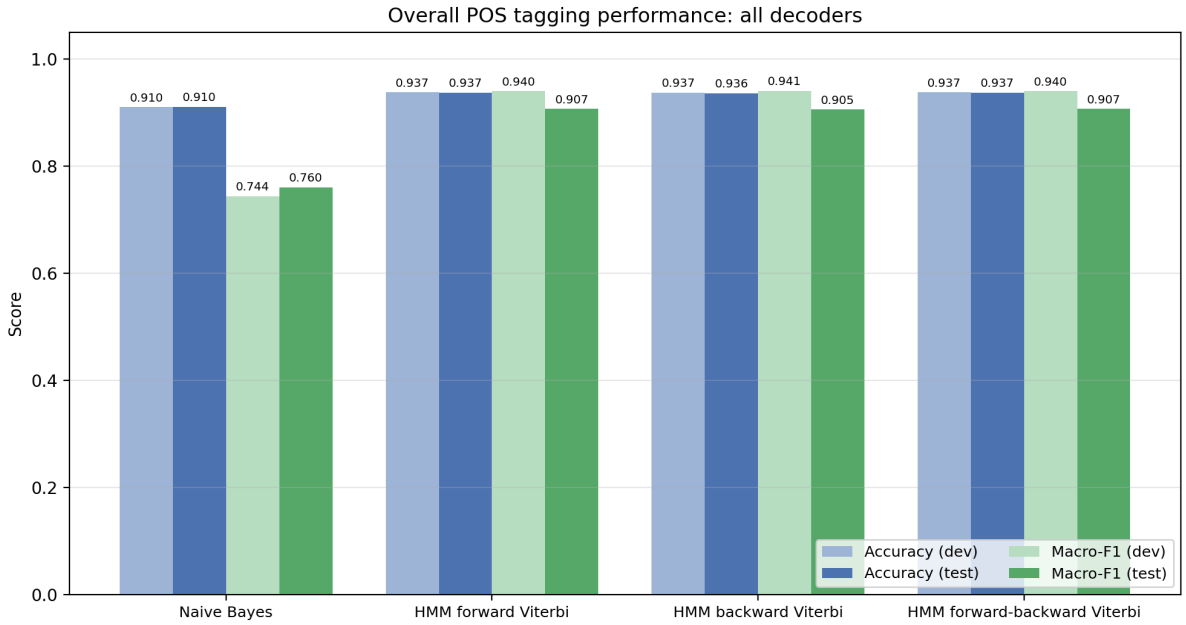
HMM. A first-order HMM with add-one smoothed transitions, MLE emissions, and words seen once in training mapped to <UNK>. Three decoders are compared: **forward Viterbi** (standard left-to-right dynamic program); **backward Viterbi** (right-to-left dynamic program over a reverse transition model P(t_i | t_{i+1}) estimated from the same bigram counts); and **forward-backward Viterbi**, which runs a forward and a backward max-product pass and assigns each token the tag on the best full-sequence path through it (max-marginal decoding). Because the last decoder picks each token independently, its output need not be a single consistent tag path.

All three Viterbi decoders were checked against brute-force search over every tag sequence on short sentences: the forward and backward decoders returned the exact best sequence, and the forward-backward decoder returned the exact per-token max-marginals.

3. Overall results

Method	Dev accuracy	Dev macro-F1	Test accuracy	Test macro-F1
Naive Bayes	0.9102	0.7436	0.9104	0.7600
HMM forward Viterbi	0.9375	0.9400	0.9372	0.9067
HMM backward Viterbi	0.9372	0.9406	0.9363	0.9055
HMM forward-backward Viterbi	0.9375	0.9400	0.9372	0.9067

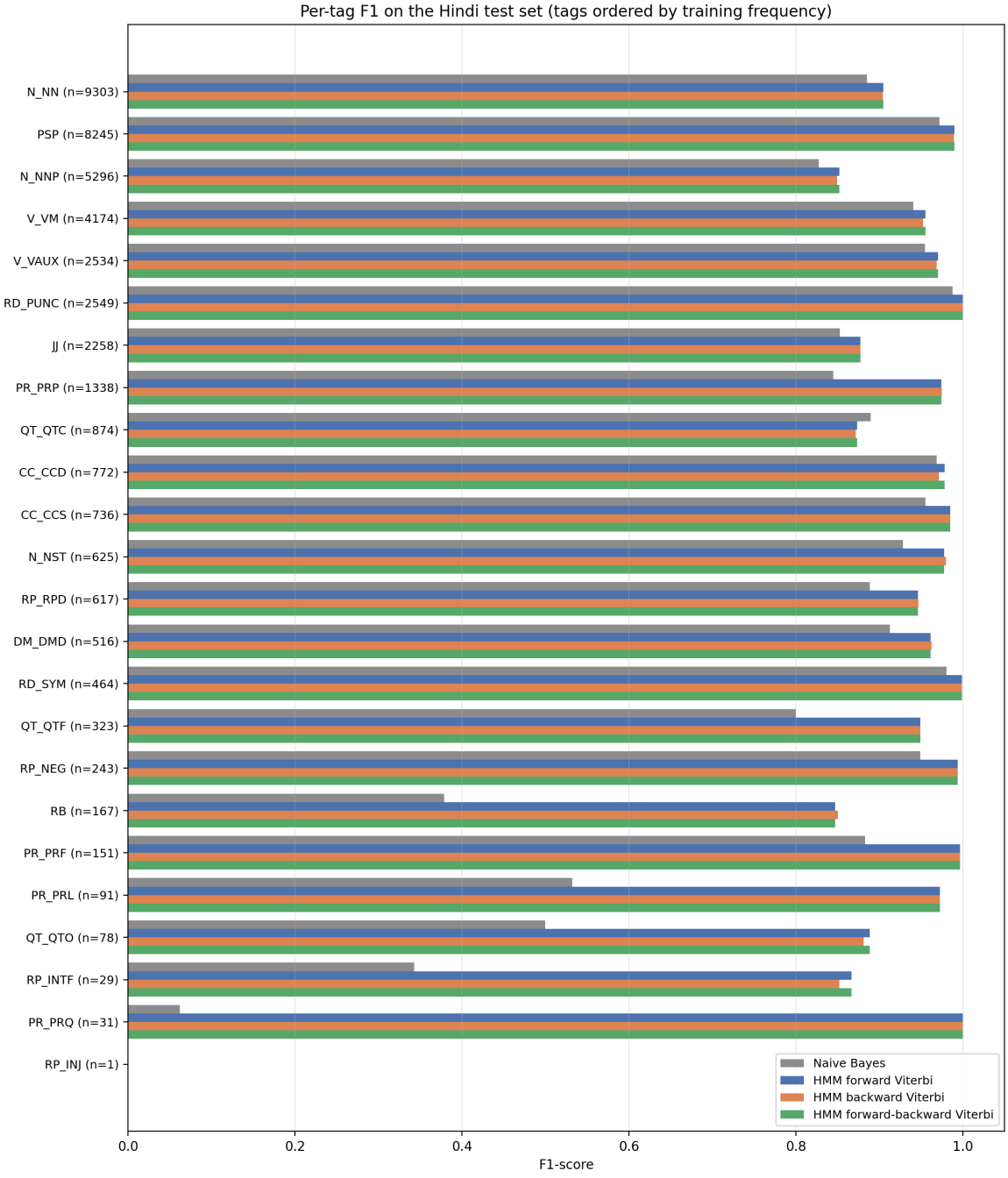
Macro-F1 averages per-tag F1 over tags that occur in the gold data.


Overall POS tagging performance: all decoders
Score
1.0
0.8
0.6
0.4
0.2
0.0
0.910
0.910
0.744
0.760
0.937
0.937
0.940
0.907
0.937
0.936
0.941
0.905
0.937
0.937
0.940
0.907
Accuracy (dev)
Accuracy (test)
Macro-F1 (dev)
Macro-F1 (test)
Naive Bayes
HMM forward Viterbi
HMM backward Viterbi
HMM forward-backward Viterbi

4. Per-tag F1 on the test set

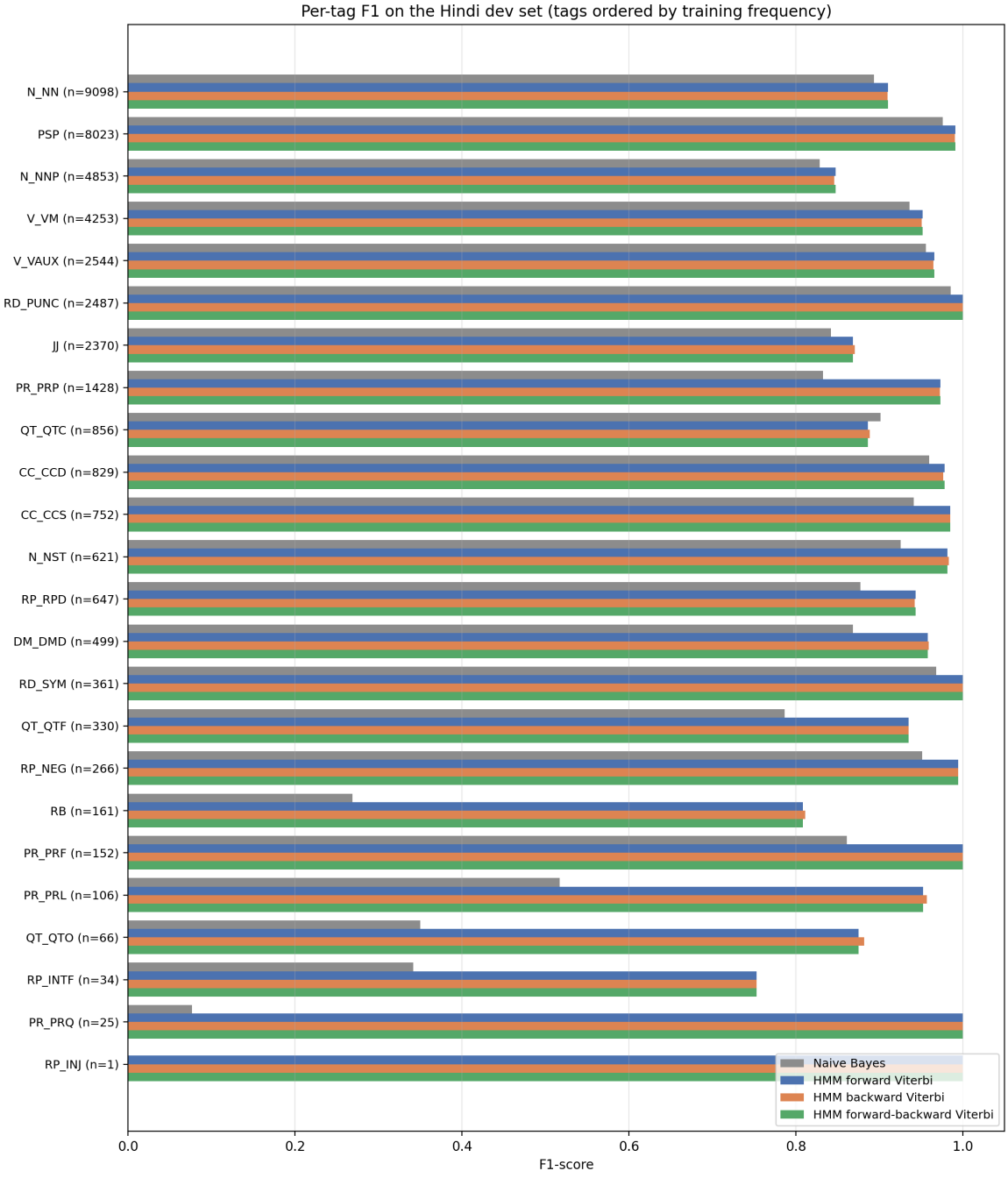
Tags are ordered by training frequency; Support is the number of gold test tokens.

Tag	Support	Naive Bayes	HMM forward Viterbi	HMM backward Viterbi	HMM forward-backward Viterbi
N_NN	9303	0.886	0.905	0.904	0.905
PSP	8245	0.972	0.990	0.989	0.990
N_NNP	5296	0.827	0.852	0.849	0.852
V_VM	4174	0.941	0.955	0.953	0.955
V_VAUX	2534	0.955	0.970	0.969	0.970
RD_PUNC	2549	0.988	1.000	1.000	1.000
JJ	2258	0.853	0.877	0.878	0.877
PR_PRP	1338	0.845	0.974	0.975	0.974
QT_QTC	874	0.890	0.874	0.872	0.874
CC_CCD	772	0.969	0.978	0.972	0.978
CC_CCS	736	0.955	0.985	0.985	0.985
N_NST	625	0.929	0.978	0.980	0.978
RP_RPD	617	0.889	0.946	0.947	0.946
DM_DMD	516	0.913	0.962	0.963	0.962
RD_SYM	464	0.981	0.999	0.999	0.999
QT_QTF	323	0.800	0.949	0.949	0.949
RP_NEG	243	0.949	0.994	0.994	0.994
RB	167	0.379	0.847	0.851	0.847
PR_PRF	151	0.883	0.997	0.997	0.997
PR_PRL	91	0.532	0.973	0.973	0.973
QT_QTO	78	0.500	0.889	0.881	0.889
RP_INTF	29	0.343	0.867	0.852	0.867
PR_PRQ	31	0.062	1.000	1.000	1.000
RP_INJ	1	0.000	0.000	0.000	0.000


Per-tag F1 on the Hindi test set (tags ordered by training frequency)
N_NN (n=9303)
PSP (n=8245)
N_NNP (n=5296)
V_VM (n=4174)
V_VAUX (n=2534)
RD_PUNC (n=2549)
JJ (n=2258)
PR_PRP (n=1338)
QT_QTC (n=874)
CC_CCD (n=772)
CC_CCS (n=736)
N_NST (n=625)
RP_RPD (n=617)
DM_DMD (n=516)
RD_SYM (n=464)
QT_QTF (n=323)
RP_NEG (n=243)
RB (n=167)
PR_PRF (n=151)
PR_PRL (n=91)
QT_QTO (n=78)
RP_INTF (n=29)
PR_PRQ (n=31)
RP_INJ (n=1)
Naive Bayes
HMM forward Viterbi
HMM backward Viterbi
HMM forward-backward Viterbi
0.0
0.2
0.4
0.6
0.8
1.0
F1-score

5. Per-tag F1 on the dev set

Tag	Support	Naive Bayes	HMM forward Viterbi	HMM backward Viterbi	HMM forward-backward Viterbi
N_NN	9098	0.893	0.910	0.910	0.910
PSP	8023	0.976	0.991	0.991	0.991
N_NNP	4853	0.829	0.848	0.846	0.848
V_VM	4253	0.936	0.952	0.951	0.952
V_VAUX	2544	0.956	0.966	0.965	0.966
RD_PUNC	2487	0.986	1.000	1.000	1.000
JJ	2370	0.842	0.868	0.870	0.868
PR_PRP	1428	0.833	0.973	0.973	0.973
QT_QTC	856	0.902	0.886	0.889	0.886
CC_CCD	829	0.960	0.978	0.977	0.978
CC_CCS	752	0.941	0.985	0.985	0.985
N_NST	621	0.926	0.982	0.983	0.982
RP_RPD	647	0.877	0.943	0.943	0.943
DM_DMD	499	0.868	0.958	0.959	0.958
RD_SYM	361	0.968	1.000	1.000	1.000
QT_QTF	330	0.786	0.935	0.935	0.935
RP_NEG	266	0.951	0.994	0.994	0.994
RB	161	0.269	0.809	0.811	0.809
PR_PRF	152	0.861	1.000	1.000	1.000
PR_PRL	106	0.517	0.952	0.957	0.952
QT_QTO	66	0.350	0.875	0.882	0.875
RP_INTF	34	0.341	0.753	0.753	0.753
PR_PRQ	25	0.077	1.000	1.000	1.000
RP_INJ	1	0.000	1.000	1.000	1.000


Per-tag F1 on the Hindi dev set (tags ordered by training frequency)
N_NN (n=9098)
PSP (n=8023)
N_NNP (n=4853)
V_VM (n=4253)
V_VAUX (n=2544)
RD_PUNC (n=2487)
JJ (n=2370)
PR_PRP (n=1428)
QT_QTC (n=856)
CC_CCD (n=829)
CC_CCS (n=752)
N_NST (n=621)
RP_RPD (n=647)
DM_DMD (n=499)
RD_SYM (n=361)
QT_QTF (n=330)
RP_NEG (n=266)
RB (n=161)
PR_PRF (n=152)
PR_PRL (n=106)
QT_QTO (n=66)
RP_INTF (n=34)
PR_PRQ (n=25)
RP_INJ (n=1)
Naive Bayes
HMM forward Viterbi
HMM backward Viterbi
HMM forward-backward Viterbi
0.0
0.2
0.4
0.6
0.8
1.0
F1-score

6. Analysis

• The HMM beats Naive Bayes by about 2.7 accuracy points and 15 macro-F1 points on test. Naive Bayes tags each token with only the previous tag as context, so it cannot use the sentence-level tag structure that the HMM models.

• Naive Bayes fails most on rare and closed-class tags: PR_PRQ (test F1 0.06) and RP_INTF (0.34), and RB (0.38) too. The sequence model recovers most of these.

• The three HMM decoders perform almost identically. Forward and forward-backward Viterbi give the same test accuracy (0.9372). Backward Viterbi is about 0.1 points lower, because its reverse model is normalised differently even though it uses the same counts.

• The hardest tags for all methods are N_NNP (proper nouns, test F1 about 0.85 over 5,296 tokens), RB (about 0.85), and RP_INTF (about 0.87). Proper nouns are often unseen words, and RB overlaps with adverbial uses of other tags.

• RP_INJ has F1 = 0 for every method. It occurs once in the test set and no decoder labels it correctly, so one token drags the per-tag average down.